Project Report: News Text Classification

1. Objective

Build and compare three machine learning models to classify news articles into four categories - World, Sports, Business, and Sci/Tech - using the AG News dataset.

2. Dataset

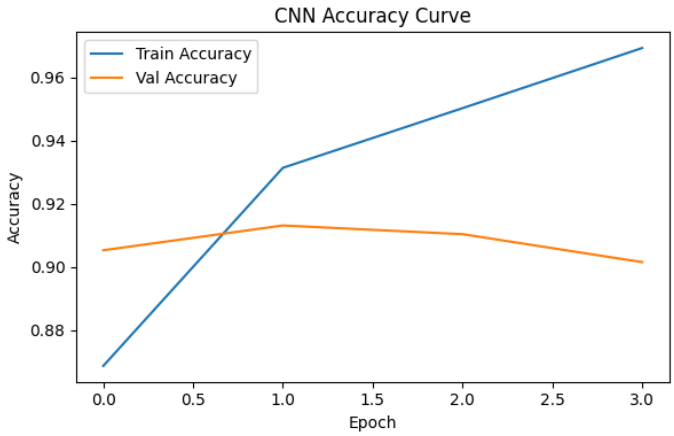
Source: AG News (fancyzhx/ag_news)
Training set: 120,000 articles (30,000 per class)
Test set: 7,600 articles (1,900 per class)
Data quality: No missing values; perfectly balanced classes

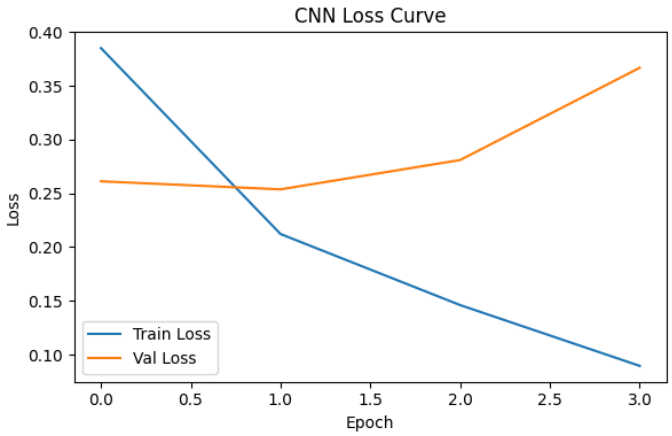
3. Methodology

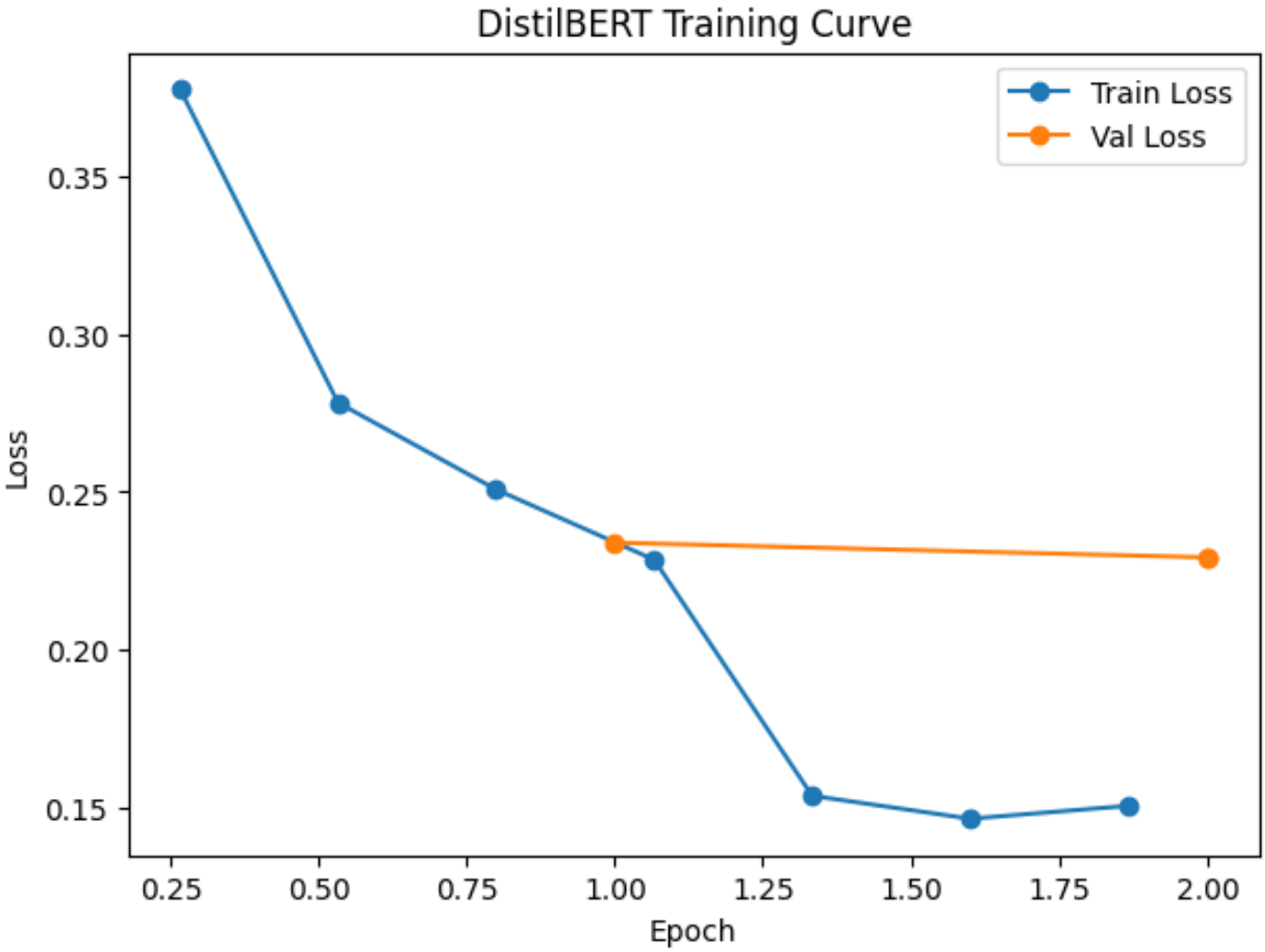
3.1 Logistic Regression - TF-IDF vectorization (10,000 features), trained with sklearn LogisticRegression.

3.2 CNN - Embedding + Conv1D + GlobalMaxPooling. Early stopping applied after initial overfitting (val accuracy improved from 89.7% to 91.14%).

3.3 DistilBERT - Fine-tuned distilbert-base-uncased on a balanced 30,000-row subset, 2 epochs, ~10 hours on CPU.





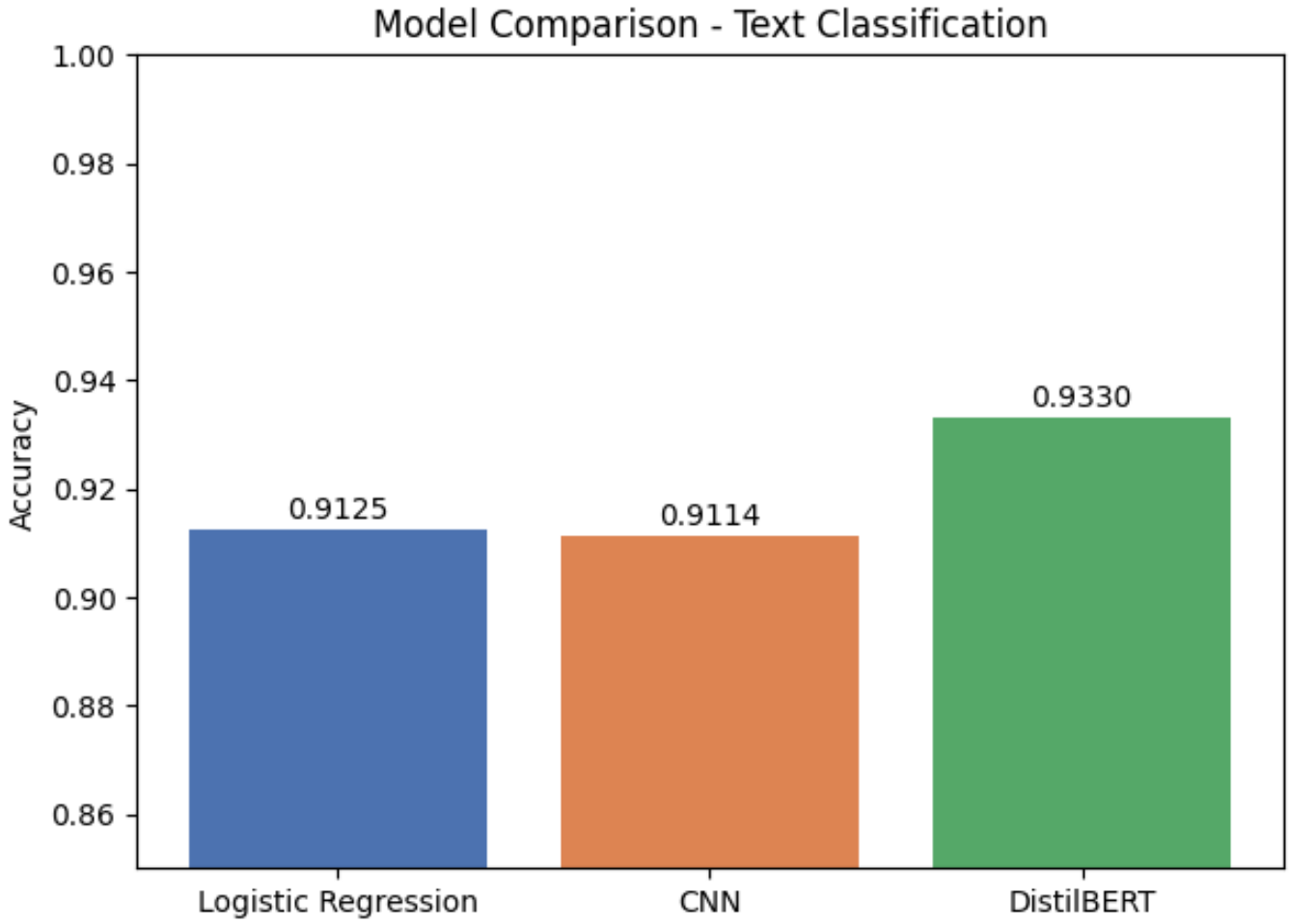


4. Results

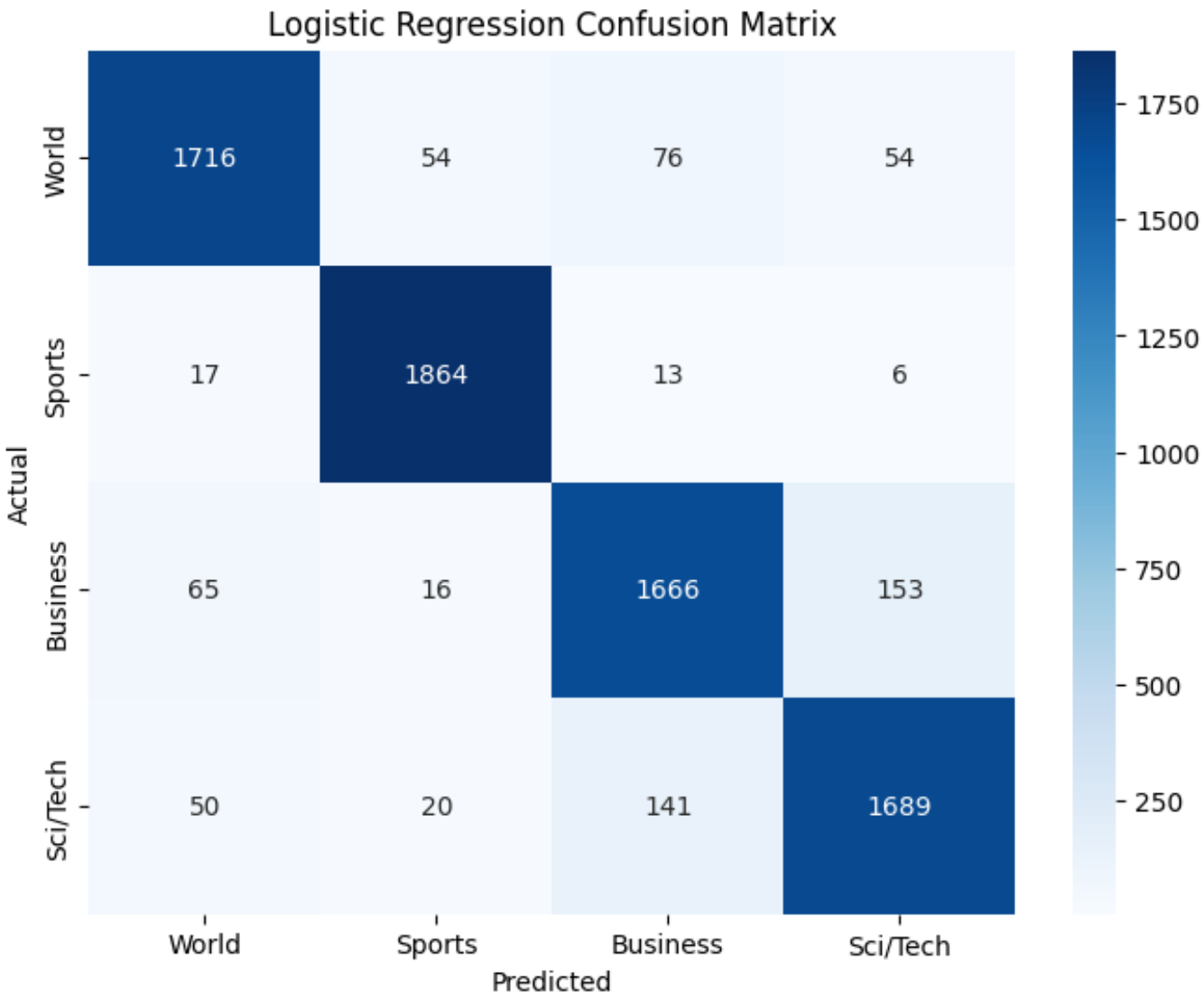
Logistic Regression: 91.25%

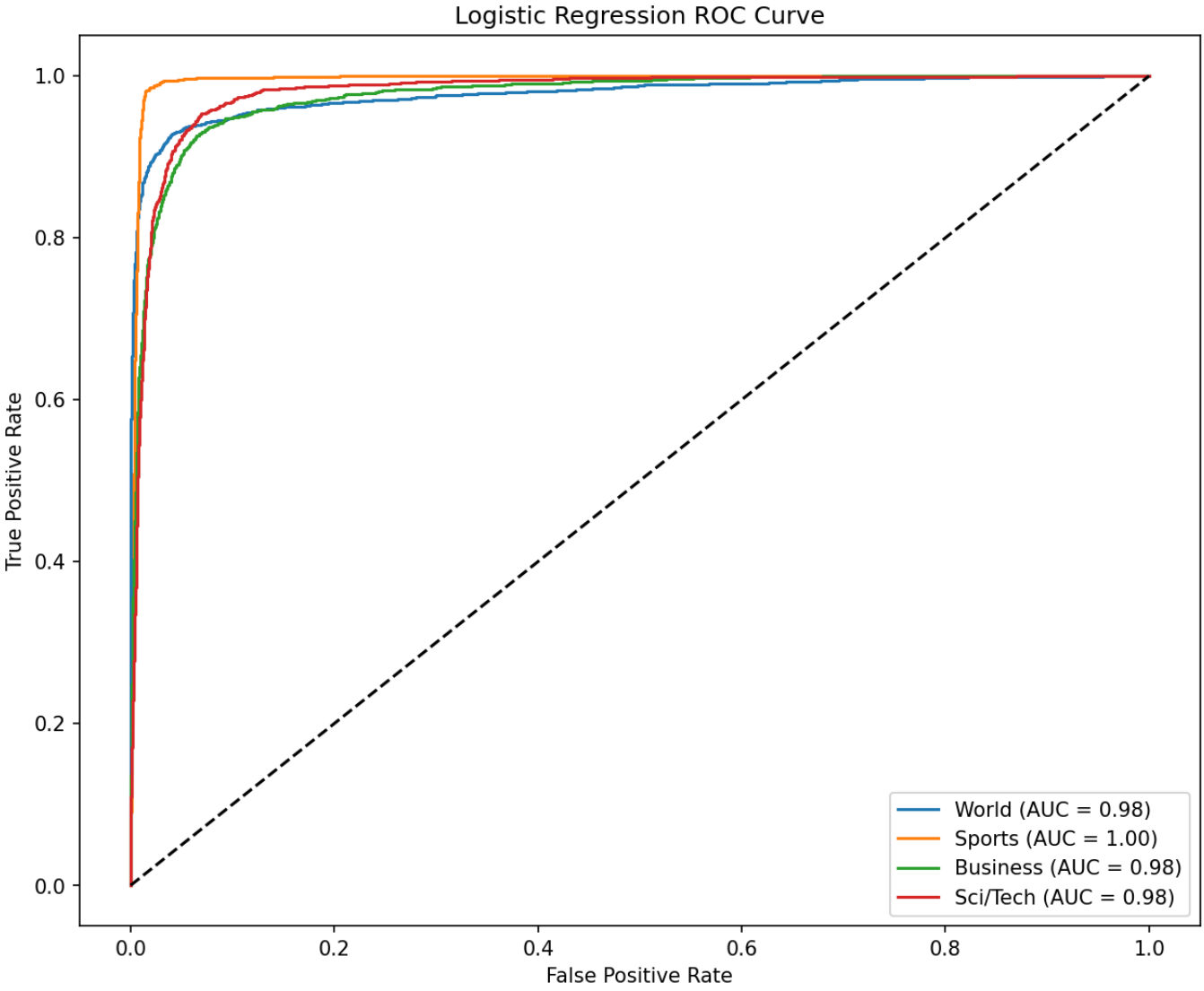
CNN: 91.14%

DistilBERT: 93.30%

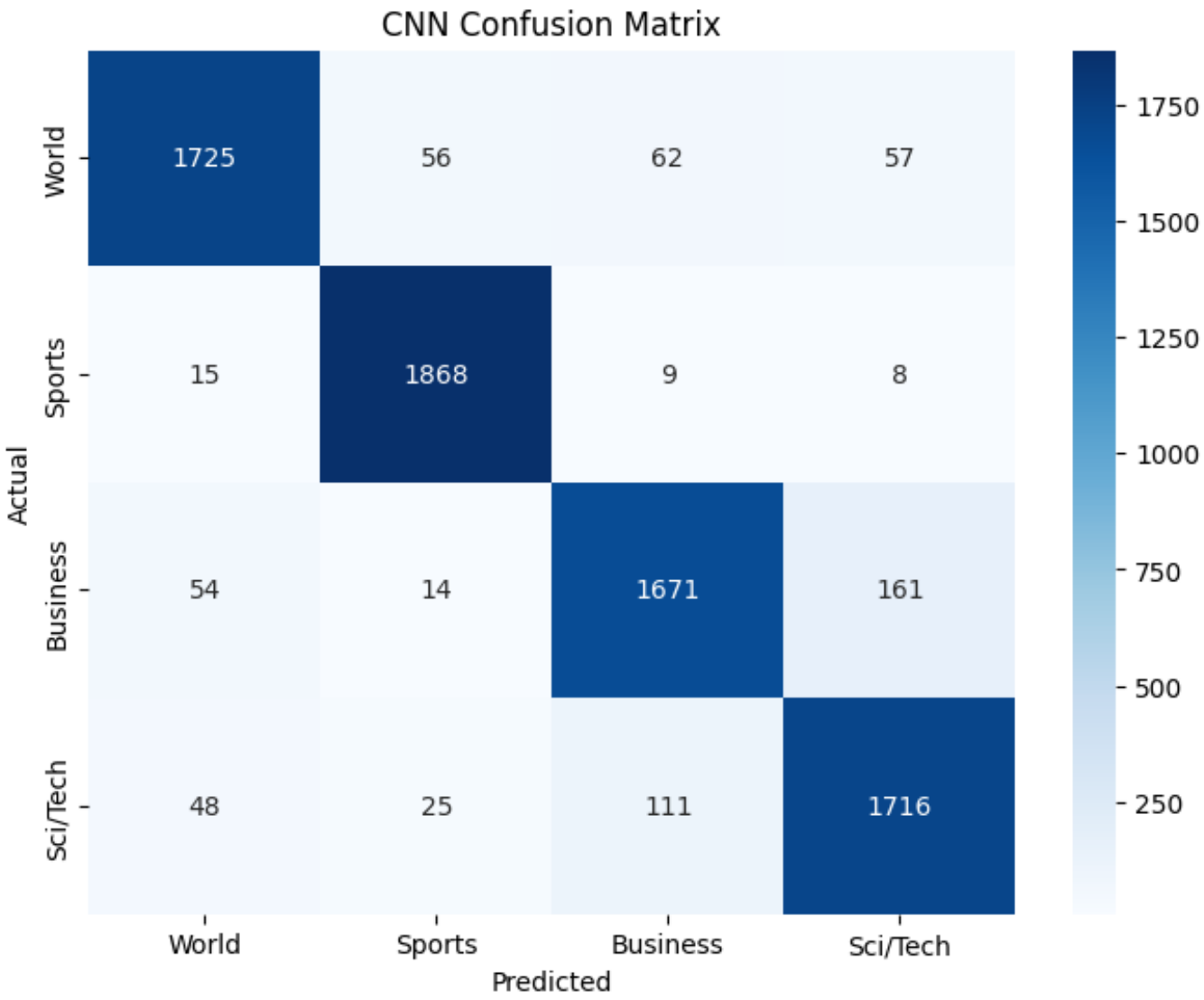


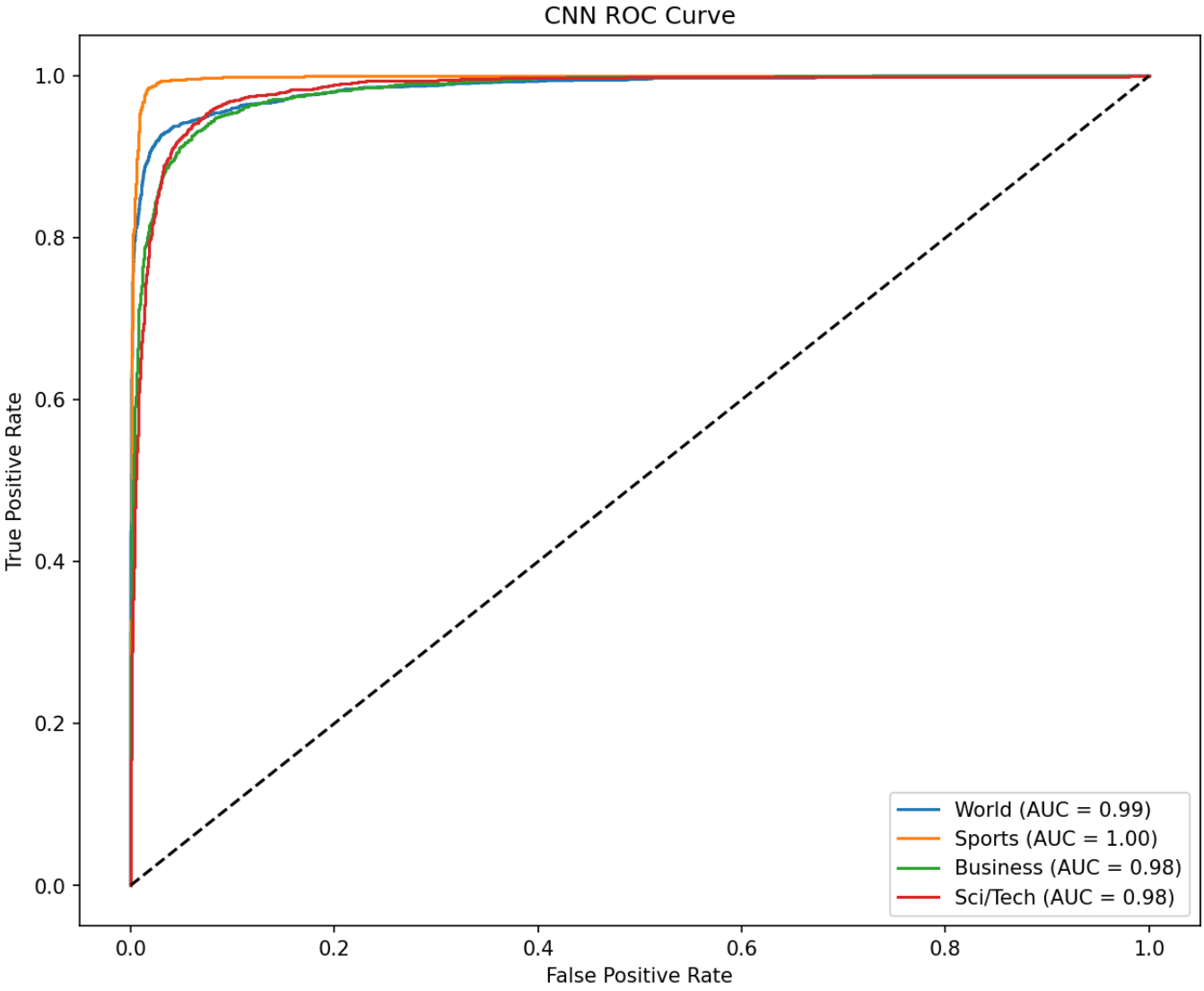
4.1 Logistic Regression


Logistic Regression Confusion Matrix
World
Sports
Business
Sci/Tech
Actual
1716
54
76
54
17
1864
13
6
65
16
1666
153
50
20
141
1689
World
Sports
Business
Sci/Tech
Predicted
1750
1500
1250
1000
750
500
250

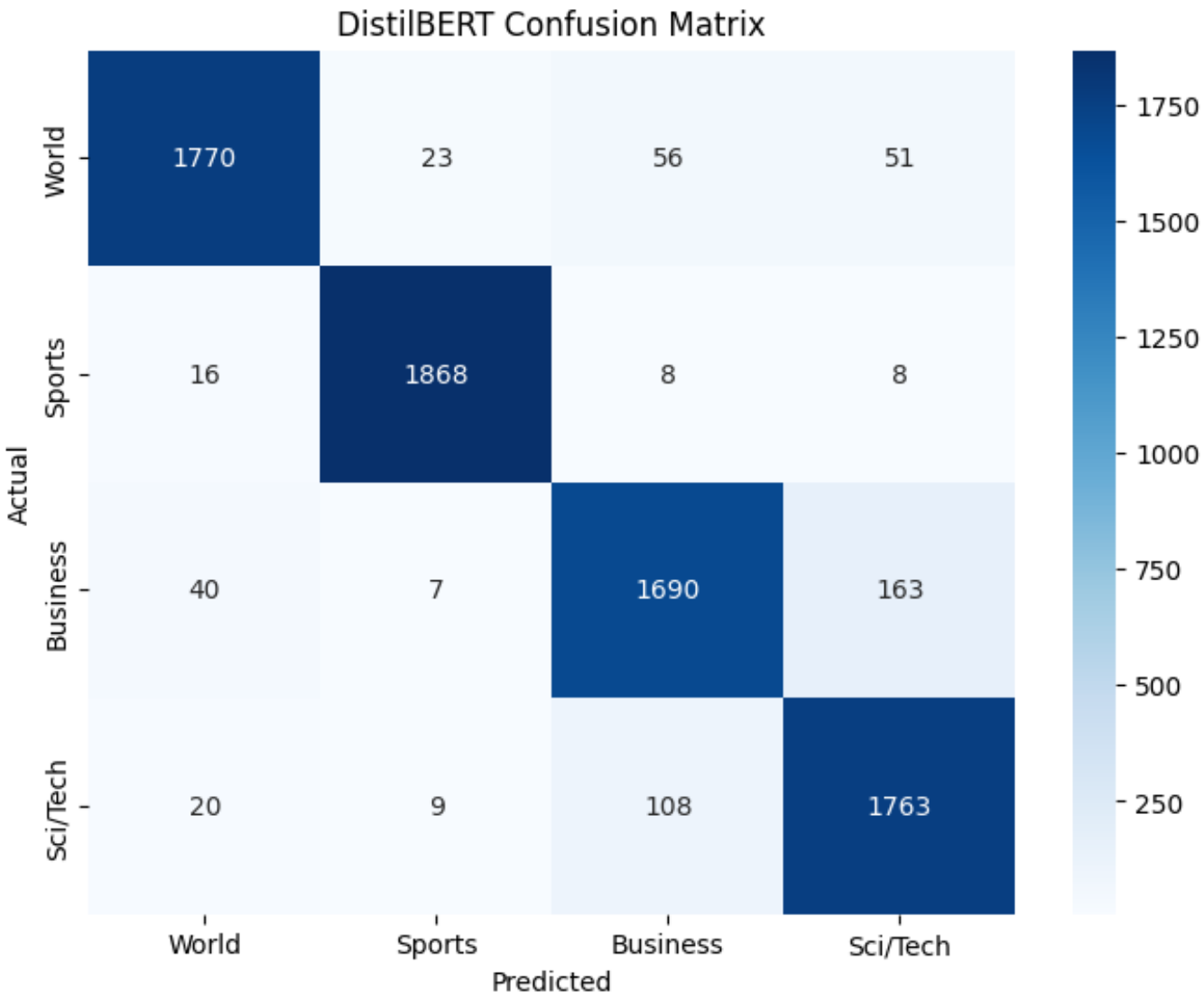

Logistic Regression ROC Curve
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
World (AUC = 0.98)
Sports (AUC = 1.00)
Business (AUC = 0.98)
Sci/Tech (AUC = 0.98)

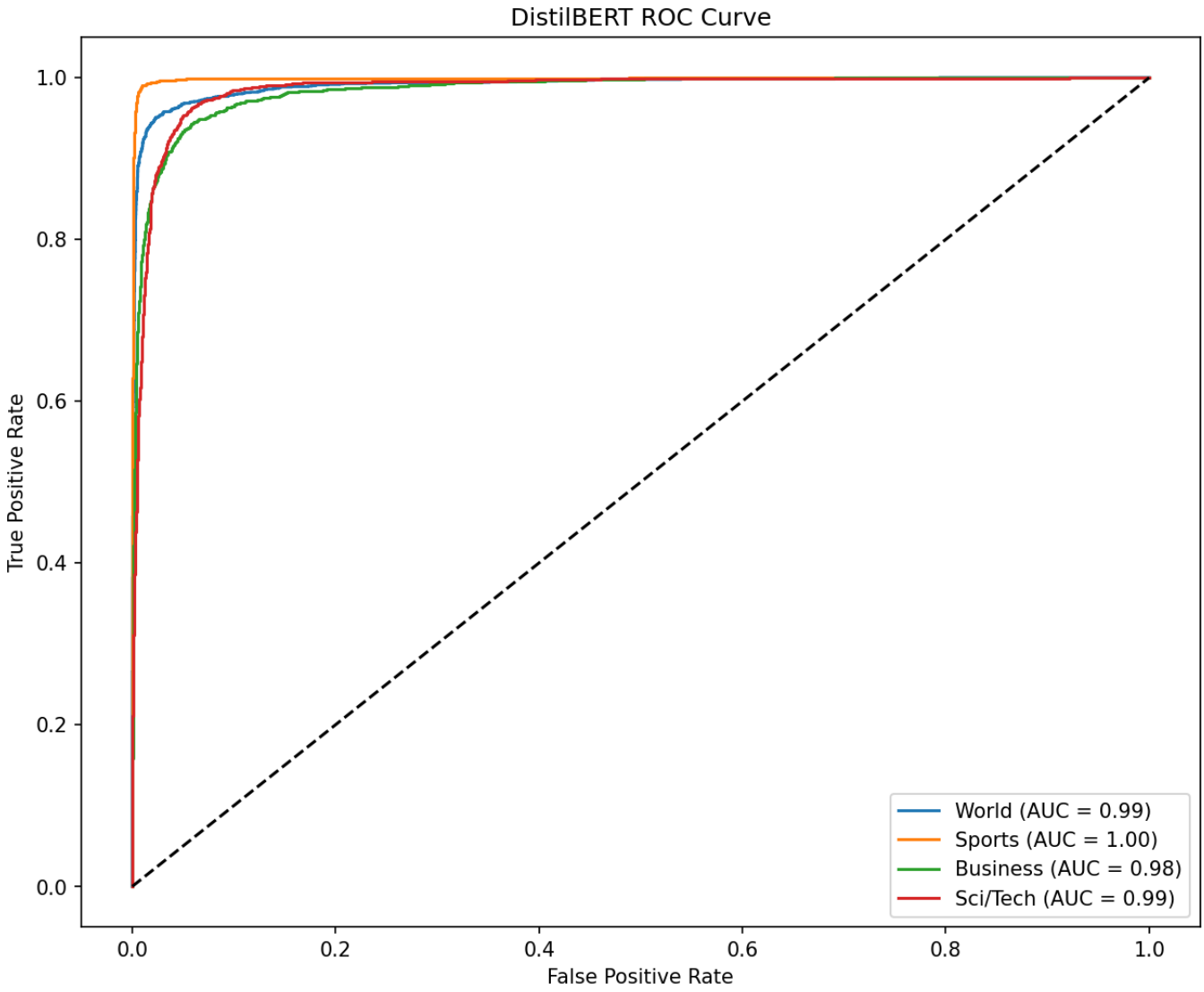
4.2 CNN


CNN Confusion Matrix
World
Sports
Business
Sci/Tech
Actual
1725
56
62
57
15
1868
9
8
54
14
1671
161
48
25
111
1716
World
Sports
Business
Sci/Tech
Predicted
1750
1500
1250
1000
750
500
250



4.3 DistilBERT


DistilBERT Confusion Matrix
World
Sports
Business
Sci/Tech
Actual
1770
23
56
51
16
1868
8
8
40
7
1690
163
20
9
108
1763
World
Sports
Business
Sci/Tech
Predicted
1750
1500
1250
1000
750
500
250



5. Discussion

All three models performed well above a random baseline (25%), with DistilBERT giving the strongest overall result. The CNN's early overfitting highlights the importance of validation monitoring. DistilBERT's advantage came from contextual understanding of language. Business and Sci/Tech remained the hardest classes to separate across all models.

6. Conclusion

DistilBERT achieved the best accuracy (93.30%) and is the recommended model for deployment. Logistic Regression remains a strong, lightweight baseline with far less computational cost.

7. Future Work

- Train DistilBERT on the full dataset with GPU access
- Experiment with ensembling predictions
- Deploy the best-performing model as a live web app